

Assignment 3

Financial Econometrics: ECO764

Parliamentary Election Impact on Indian Capital Markets

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Question 1: Options Strategies for May 15, 2009

May 15, 2009 was a tense trading session. With parliamentary vote counting set to begin the very next morning, Raj Sharma had to decide on a set of options strategies for his clients before the market closed. The Nifty was trading at 3,671.65 and, depending on which party came to power, the index could either surge sharply or tumble. Sharma settled on four strategies drawn from Exhibit 3 data -- two built around a likely bull outcome and two designed to profit from extreme movement in either direction.

1.1 Why these strategies?

Most pre-election surveys pointed toward an INC-led UPA returning with a workable majority, which is why the primary lean was bullish. Indian elections don't always go the way polls suggest though -- the 2004 shock wiped around 20% off Nifty in just two sessions. So alongside the bullish bets, Sharma wanted something that paid off if the index cratered instead. The straddle and strangle handled that without needing a firm directional view.

1.2 Strategy Details (from Exhibit 3)

#	Strategy	Type	Components	Max Loss	Max Profit
1	Long Call	Bull / Aggressive	Buy Call K=3800 @ Rs.108	Rs.108	Unlimited
2	Bull Call Spread	Bull / Defensive	Buy Call K=3700 @ Rs.150 Sell Call K=3900 @ Rs.70 Net cost = Rs.80	Rs.80	Rs.120
3	Long Straddle	Volatile	Buy Call K=3700 @ Rs.150 Buy Put K=3700 @ Rs.162 Total cost = Rs.312	Rs.312	Unlimited
4	Long Strangle	Volatile / Cheaper	Buy Call K=3800 @ Rs.108 Buy Put K=3600 @ Rs.119.65 Total = Rs.227.65	Rs.227.65	Unlimited

1.3 Payoff Formulas

Long Call : $\text{Max}(S_T - 3800, 0) - 108$

Bull Call Spread: $\text{Max}(S_T - 3700, 0) - \text{Max}(S_T - 3900, 0) - 80$

Long Straddle : $\text{Max}(S_T - 3700, 0) + \text{Max}(3700 - S_T, 0) - 312$

Long Strangle : $\text{Max}(S_T - 3800, 0) + \text{Max}(3600 - S_T, 0) - 227.65$

1.4 Payoff Diagrams

Q1: Options Strategies | May 15, 2009 | Nifty Spot = 3671.65

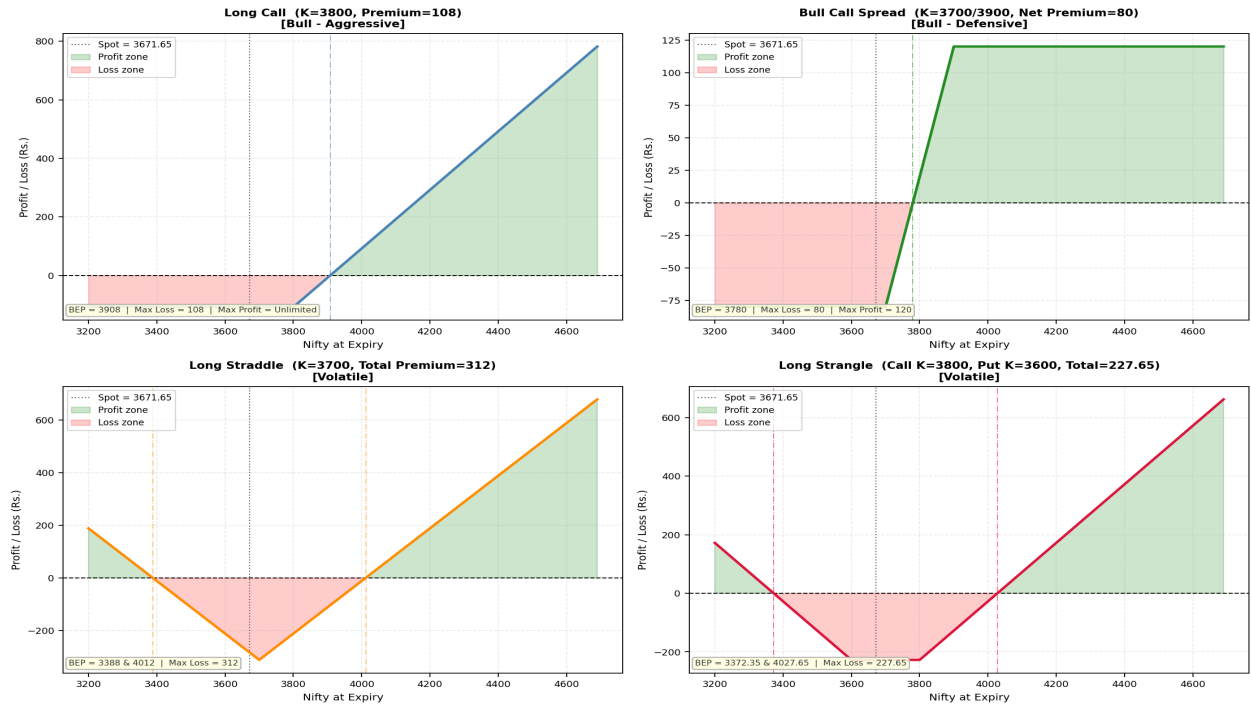


Figure 1: Payoff at expiry for all four strategies. Green shading = profit zone, red shading = loss zone, dotted grey line = Nifty spot on May 15 (3,671.65).

Question 2: Performance Evaluation of the Strategies

Monday morning, May 18, turned out to be one of the most dramatic openings Indian markets had ever seen. The INC had crossed 200 seats -- far better than most polls had forecast -- and the SGX Nifty was already up over 11% before the NSE even opened. When trading started at 9:50 a.m., the upper circuit kicked in within three seconds. All four strategies Sharma had recommended were sitting on massive unrealised gains, but nobody could book them because the market was frozen. Sharma advised clients to sell as soon as the halt lifted, locking in 75-80% of the open profit. Tuesday's closing prices (Exhibit 6) are used here to calculate the full outcome, with Nifty finishing at 4,318.45 -- a 17.6% jump from Friday's close.

2.1 Exit Prices -- May 19, 2009 (Exhibit 6)

Option	Strike	Entry (Rs.)	Exit LTP (Rs.)	Net Change (Rs.)
Call -- Long Call	3800	108	535	+427
Call -- Buy leg (BS)	3700	150	626	+476
Call -- Sell leg (BS)	3900	70	449	-379 (cost)
Put -- Straddle	3700	162	18.85	-143.15 (loss)
Put -- Strangle	3600	119.65	14	-105.65 (loss)

2.2 Profit and Return Summary

Strategy	Entry Cost (Rs.)	Net Profit (Rs.)	Return (%)
Long Call (K=3800)	108	427	395.4%
Bull Call Spread (3700/3900)	80	97	121.2%
Long Straddle (K=3700)	312	332.85	106.7%
Long Strangle (3800C/3600P)	227.65	321.35	141.2%

Q2: Strategy Performance | May 15 to May 19, 2009 | Nifty: 3671.65 -> 4318.45 (+17.6%)

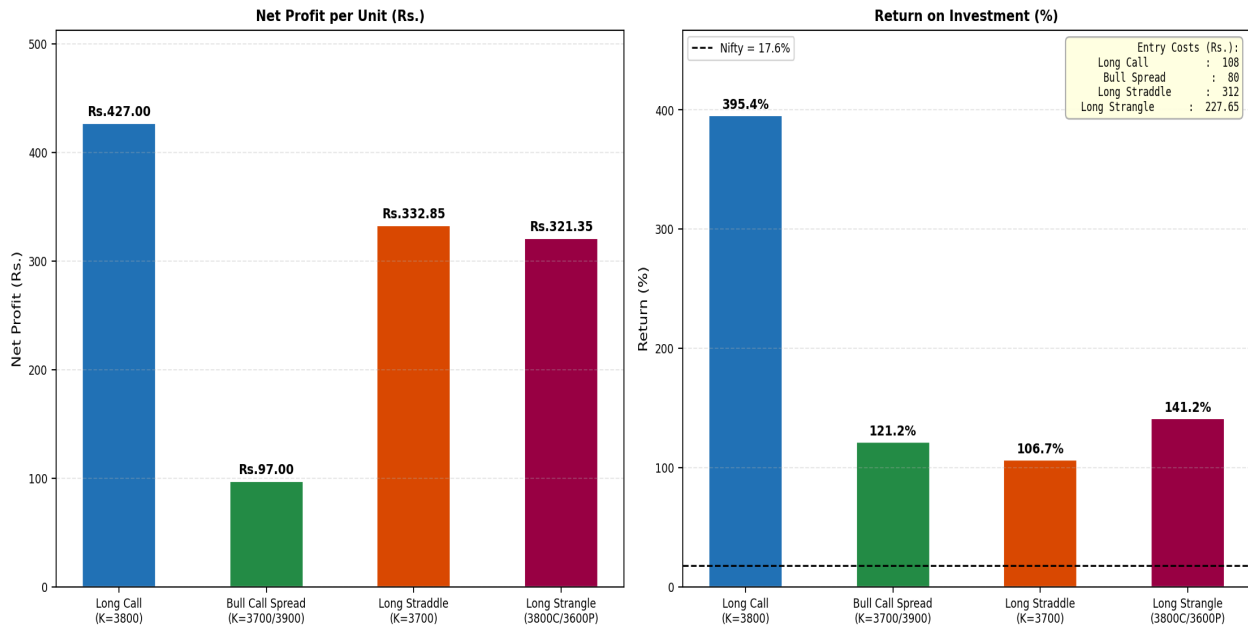


Figure 2: Left panel shows absolute profit per unit; right panel shows percentage return on each strategy. The dashed line marks the Nifty index return of 17.6%.

2.3 What the numbers tell us

All four strategies closed in the green, which reflects just how one-sided the market move was. A few numbers are worth looking at closely:

- **Long Call turned Rs.108 into Rs.535.** That is a 395% return in four days. The strategy had no ceiling on the upside, and the market gave it full room to run.
- **Long Strangle finished second at 141%.** Paying Rs.227.65 for options on both sides felt steep going in, but the call leg alone brought back Rs.427. The put leg fizzled to Rs.14, almost a full write-off, but it barely scratched the total return.
- **Bull Call Spread gave back 121%** on a Rs.80 outlay. The gain was capped at Rs.120 by design and it hit that cap. For clients who prioritised limiting downside over chasing upside, this was exactly the right trade.
- **Long Straddle returned 107%** even with the put leg bleeding Rs.143. The call leg gain of Rs.476 was more than enough to absorb that and still double the investment.
- **Timing helped too.** Getting in during the last 15 minutes of May 15 meant buying into elevated implied vol right before a weekend event, and theta had no time to eat into the positions before Monday's gap open.

Question 3: Hedging Effectiveness of Stock Options

Beyond index strategies, Sharma also had clients holding individual stocks whose downside in a bad election result could be severe. Hedging effectiveness measures what fraction of that maximum possible loss a protective put eliminates -- a score of 92% means the put absorbs 92% of the worst-case fall, leaving only 8% exposed. Higher strikes raise that score but cost more premium, so the choice of strike is always a trade-off between protection and the drag on returns.

3.0 Hedge Effectiveness Formula

Hedging Effectiveness measures what fraction of the maximum possible downside loss is eliminated by holding the put option:

$$\text{Hedge Effectiveness (\%)} = [1 - (\text{Max Loss WITH Hedge} / \text{Max Loss WITHOUT Hedge})] \times 100$$

Where:

Max Loss WITHOUT Hedge = Entry Stock Price (worst case: stock falls to zero)

Max Loss WITH Hedge = (Spot_entry - Strike) + Premium [if Strike < Spot_entry, OTM put]
= Premium alone [if Strike >= Spot_entry, ATM/ITM put]

Effective Floor Price = Strike - Premium (guaranteed minimum exit price)

3.1 ICICI Bank

ICICI Bank was trading at Rs.574.70 on May 15 and closed at Rs.756.15 on May 19, a gain of Rs.181.45 (+31.6%). Five put options were available. The K=560 strike (slightly OTM) was originally chosen; the table below compares all five.

Strike (K)	Entry Premium (Rs.)	Exit LTP (Rs.)	Net Put P&L; (Rs.)	Hedged Gain (Rs.)	Effective Floor (Rs.)	Net Cost (% of gain)	Hedge Effectiveness (%)
500	13.40	1.85	-11.55	169.90	486.60	6.37%	84.67%
510	14.75	1.65	-13.10	168.35	495.25	7.22%	86.18%
520	17.00	2.45	-14.55	166.90	503.00	8.02%	87.52%
540	23.00	2.45	-20.55	160.90	517.00	11.33%	89.96%
560 *	29.10	2.50	-26.60	154.85	530.90	14.66%	92.38%

Table 1: ICICI Bank — all five available put strikes. * = strike selected in original strategy. Red = put P&L; (loss on put leg since market rose); Green = positive outcome columns. Unhedged stock gain = Rs.181.45.

- **K=500 (Deep OTM, Rs.13.40 premium)** Cheapest hedging option, net cost only Rs.11.55 (6.37% of gain). Floor at Rs.486.60. Effectiveness = 84.67% -- the put would have absorbed 84.7% of the worst-case loss in a crash scenario.
- **K=540 (OTM, Rs.23.00 premium):** A reasonable middle ground. Floor moves up to Rs.517, effectiveness reaches 89.96%, at a net cost of Rs.20.55 (11.33% of gain).
- **K=560 (Slightly OTM, Rs.29.10 premium) -- CHOSEN:** Highest protection available. Floor at Rs.530.90, effectiveness = 92.38%. Net cost Rs.26.60 (14.66% of gain). Only Rs.43.80 below entry, so even a small fall would trigger the put.

3.2 Reliance Industries

Reliance was at Rs.1,950.70 on May 15 and closed at Rs.2,230.90 on May 19, a gain of Rs.280.20 (+14.4%). Five put strikes were available ranging from deep OTM (K=1830) to near ATM (K=1950). The K=1920 strike was originally chosen.

Strike (K)	Entry Premium (Rs.)	Exit LTP (Rs.)	Net Put P&L; (Rs.)	Hedged Gain (Rs.)	Effective Floor (Rs.)	Net Cost (% of gain)	Hedge Effectiveness (%)
1830	40.50	5.00	-35.50	244.70	1789.50	12.67%	91.74%
1860	48.00	6.15	-41.85	238.35	1812.00	14.94%	92.89%
1890	56.10	8.95	-47.15	233.05	1833.90	16.83%	94.01%
1920 *	71.00	12.25	-58.75	221.45	1849.00	20.97%	94.79%
1950	84.00	15.75	-68.25	211.95	1866.00	24.36%	95.66%

Table 2: Reliance Industries — all five available put strikes. * = strike selected in original strategy.

Unhedged stock gain = Rs.280.20.

- **K=1830 (Deep OTM, Rs.40.50 premium):** Net cost = Rs.35.50 (12.67% of gain). Floor at Rs.1,789.50, effectiveness = 91.74%.
- **K=1920 (Slightly OTM, Rs.71.00 premium) -- CHOSEN:** Floor at Rs.1,849, effectiveness = 94.79%. Net cost = Rs.58.75 (20.97% of gain after recovering Rs.12.25 at exit). Only Rs.30.70 below entry -- strong near-term protection.
- **K=1950 (Near ATM, Rs.84.00 premium):** Highest floor at Rs.1,866, effectiveness = 95.66%. Most expensive but tightest protection -- any decline from entry immediately triggers the put.

3.3 Payoff Chart for Originally Chosen Strikes

Q3: Hedging Effectiveness – Protective Put Strategy

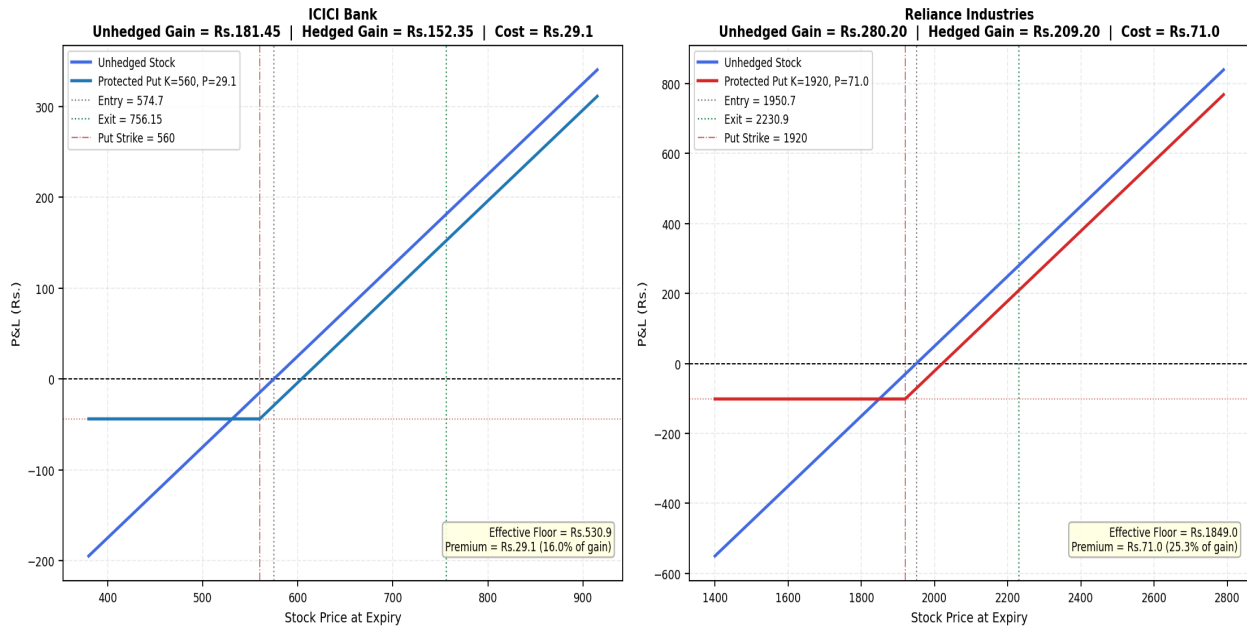


Figure 3: Protective Put payoff curves for ICICI Bank (K=560, left) and Reliance Industries (K=1920, right). Blue line = unhedged stock; coloured line = hedged position. Dotted lines mark entry price, exit price, and put strike.

3.4 Hedge Effectiveness vs Strike Price

The chart below plots hedging effectiveness (%) for every available put strike for both stocks. The annotation inside each bar shows the net hedging cost as a percentage of the stock gain -- this is the price paid for protection.

Hedge Effectiveness (%) vs Put Strike Price
ICICI Bank and Reliance Industries — May 15, 2009
Effectiveness = $[1 - (\text{Max Loss WITH Hedge}) \div (\text{Max Loss WITHOUT Hedge})] \times 100$
White text inside bars = Net Hedge Cost as % of Stock Gain

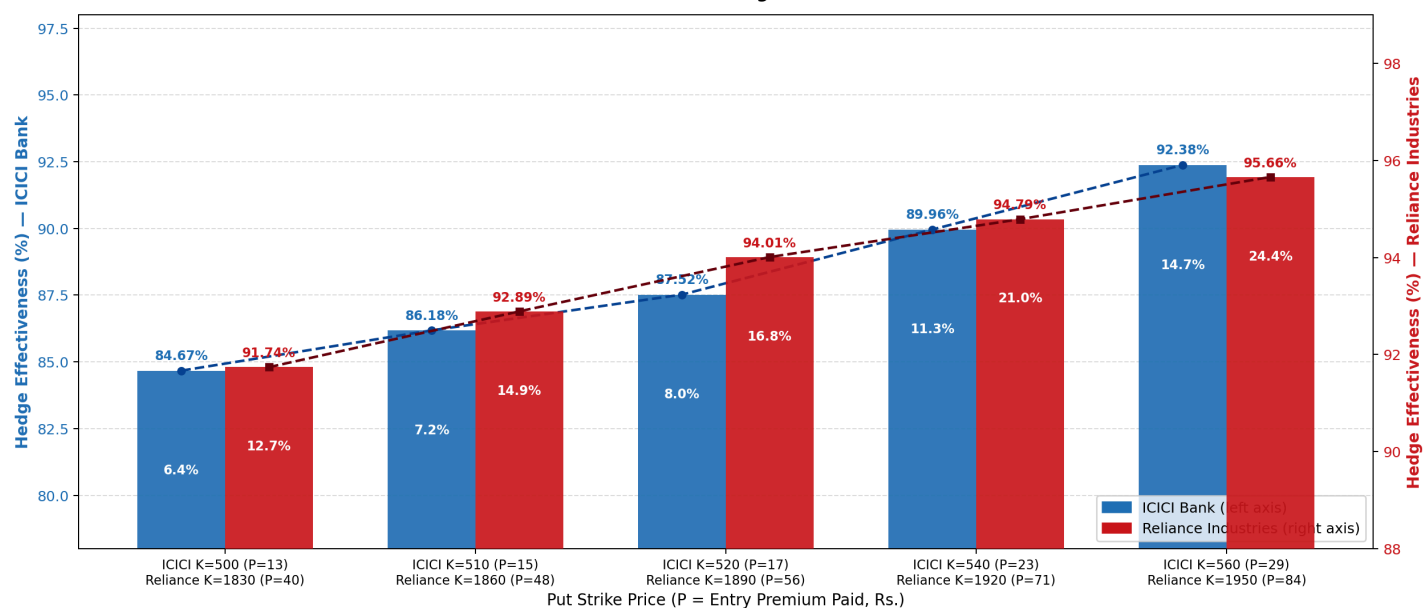


Figure 4: Hedge Effectiveness (%) vs Put Strike Price. Blue bars = ICICI Bank (left axis); Red bars = Reliance Industries (right axis). White text inside bars = net hedge cost as % of stock gain. Trend lines show the increasing effectiveness as strike rises toward spot price.

3.5 Payoff Curves Across All Available Strikes

Each coloured line below shows the P&L profile for a different strike. Darker shades represent higher strikes. The vertical dotted lines mark each strike's effective floor. The divergence between hedged and unhedged lines on the left side of each chart shows exactly how much downside each strike protects against.

Protective Put Payoff Curves – All Available Strike Prices
ICICI Bank (left) | Reliance Industries (right)

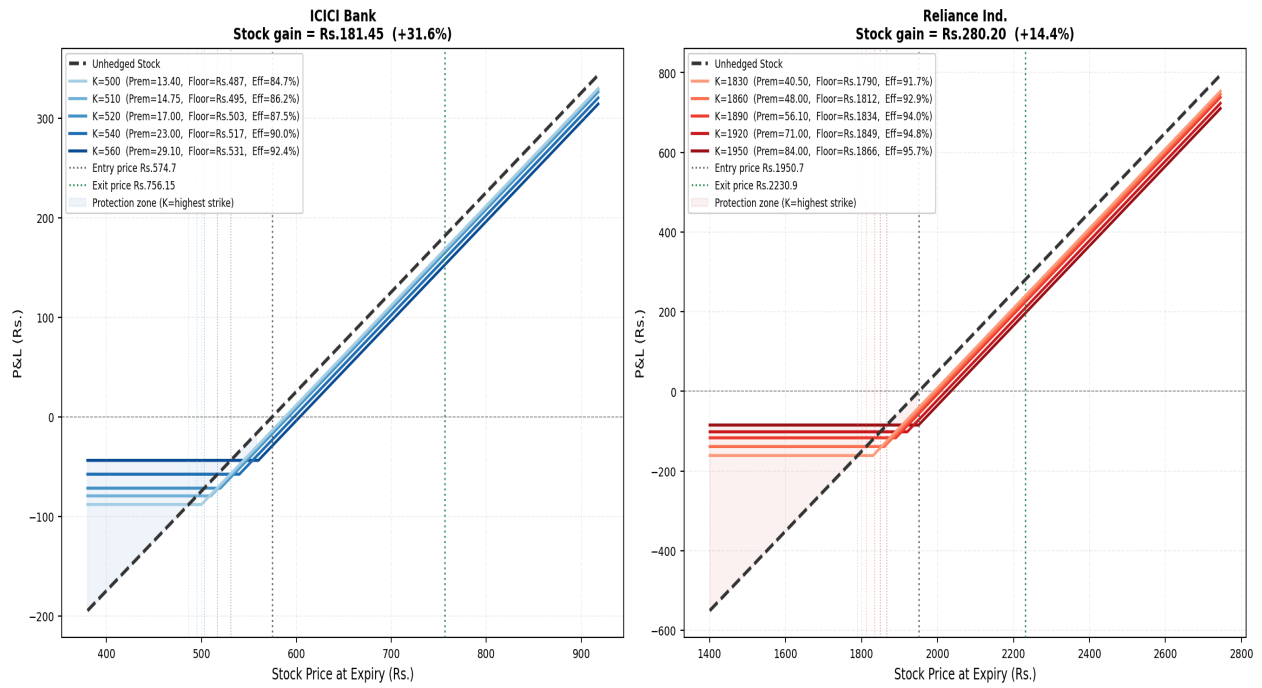


Figure 5: Protective Put payoff curves for all available strikes. ICICI Bank (left), Reliance Industries (right). Dashed grey = unhedged stock. Coloured lines (darker = higher strike). Vertical dotted lines = effective floor for each strike.

3.6 Comprehensive Comparison: Cost vs Effectiveness vs Floor

The four-panel chart below brings all three dimensions together for each stock. Top panels compare hedged gain (Rs.) across strikes -- bars below the dashed unhedged line represent the cost of protection. Bottom panels compare hedging effectiveness (green) against net hedging cost (orange), with the effective floor price plotted as a secondary axis.

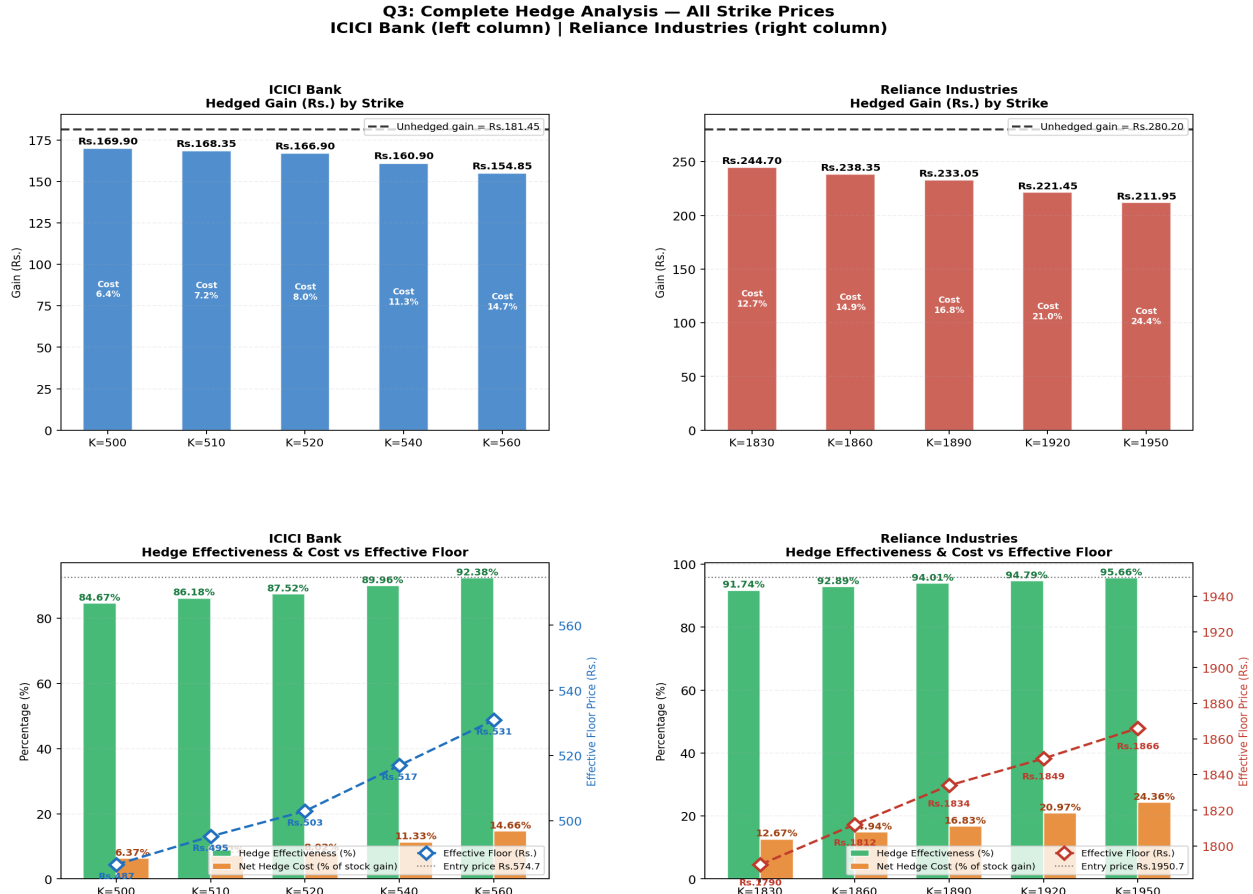


Figure 6: Four-panel hedge analysis. Top: Hedged Gain (Rs.) by strike -- dashed line = unhedged benchmark. Bottom: Green = Hedge Effectiveness (%); Orange = Net Hedge Cost (% of gain); Diamond line (right axis) = Effective Floor Price (Rs.).

3.7 Summary Comparison Table

Stock	Strike (K)	Premium (Rs.)	Hedged Gain (Rs.)	Net Cost (% of gain)	Effective Floor (Rs.)	Hedge Eff. (%)
ICICI Bank	500	13.40	169.90	6.37%	Rs.486.60	84.67%
ICICI Bank	510	14.75	168.35	7.22%	Rs.495.25	86.18%
ICICI Bank	520	17.00	166.90	8.02%	Rs.503.00	87.52%
ICICI Bank	540	23.00	160.90	11.33%	Rs.517.00	89.96%
ICICI Bank	560 *	29.10	154.85	14.66%	Rs.530.90	92.38%
Reliance	1830	40.50	244.70	12.67%	Rs.1789.50	91.74%
Reliance	1860	48.00	238.35	14.94%	Rs.1812.00	92.89%

Stock	Strike (K)	Premium (Rs.)	Hedged Gain (Rs.)	Net Cost (% of gain)	Effective Floor (Rs.)	Hedge Eff. (%)
Reliance	1890	56.10	233.05	16.83%	Rs.1833.90	94.01%
Reliance	1920 *	71.00	221.45	20.97%	Rs.1849.00	94.79%
Reliance	1950	84.00	211.95	24.36%	Rs.1866.00	95.66%

Table 3: Full summary. * = strike used in original analysis. Unhedged gains: ICICI = Rs.181.45 | Reliance = Rs.280.20.

3.8 Interpretation

Hedging effectiveness goes up as the strike moves closer to the spot price -- that much is consistent across both stocks. But every extra rupee of protection costs more premium, so there's no free lunch. Whether paying for a tighter floor makes sense really comes down to how much of a fall the investor is worried about and how much they're willing to spend.

- **OTM options are cheaper** but don't kick in until the stock falls past the strike. The K=500 ICICI put at Rs.13.40 and the K=1830 Reliance put at Rs.40.50 both gave reasonable protection (84.67% and 91.74%) at low cost, but a 5-10% fall wouldn't have triggered them quickly. Fine if you're mainly worried about a genuine crash and want to keep costs down.
- **ATM options are the practical middle ground.** With delta near -0.5 they start moving almost immediately when the stock falls. The K=560 ICICI put (92.38%, floor Rs.530.90) and K=1920 Reliance put (94.79%, floor Rs.1,849) both reflect this -- even an overnight dip of a few percent would have triggered them, keeping the maximum downside to Rs.43.80 and Rs.30.70 below entry respectively.
- **ITM options track the stock tightly** and give the highest effectiveness, but no ITM strikes were available here. The K=1950 Reliance put (95.66%) shows what happens near the ITM boundary -- paying Rs.13 more than the K=1920 only improved effectiveness by 0.87 percentage points. At some point the extra premium just isn't worth it.

Higher effectiveness doesn't automatically mean a better strategy -- the more expensive puts also eat more into returns when the market goes up, as it did here. That said, in a 2004-type scenario where Nifty dropped 20% in two days, the K=560 ICICI put would have capped losses at Rs.43.80 versus Rs.114.94 unhedged. That's a huge difference for a modest upfront cost. ATM options tend to strike the most workable balance between protection and price, which is probably why they're the go-to choice in event-driven hedging.

Question 4: Return Potential of Key Nifty 50 Stocks

Nifty was up 17.6% between May 15 and May 19, but that aggregate figure masks a lot. Some stocks gained nearly three times the index; others barely moved, and one actually fell. Going through all 47 available Nifty 50 names from Exhibit 5, the dispersion is strikingly wide -- and the pattern of who won and who didn't reflects how election outcomes feed through different sectors in very different ways.

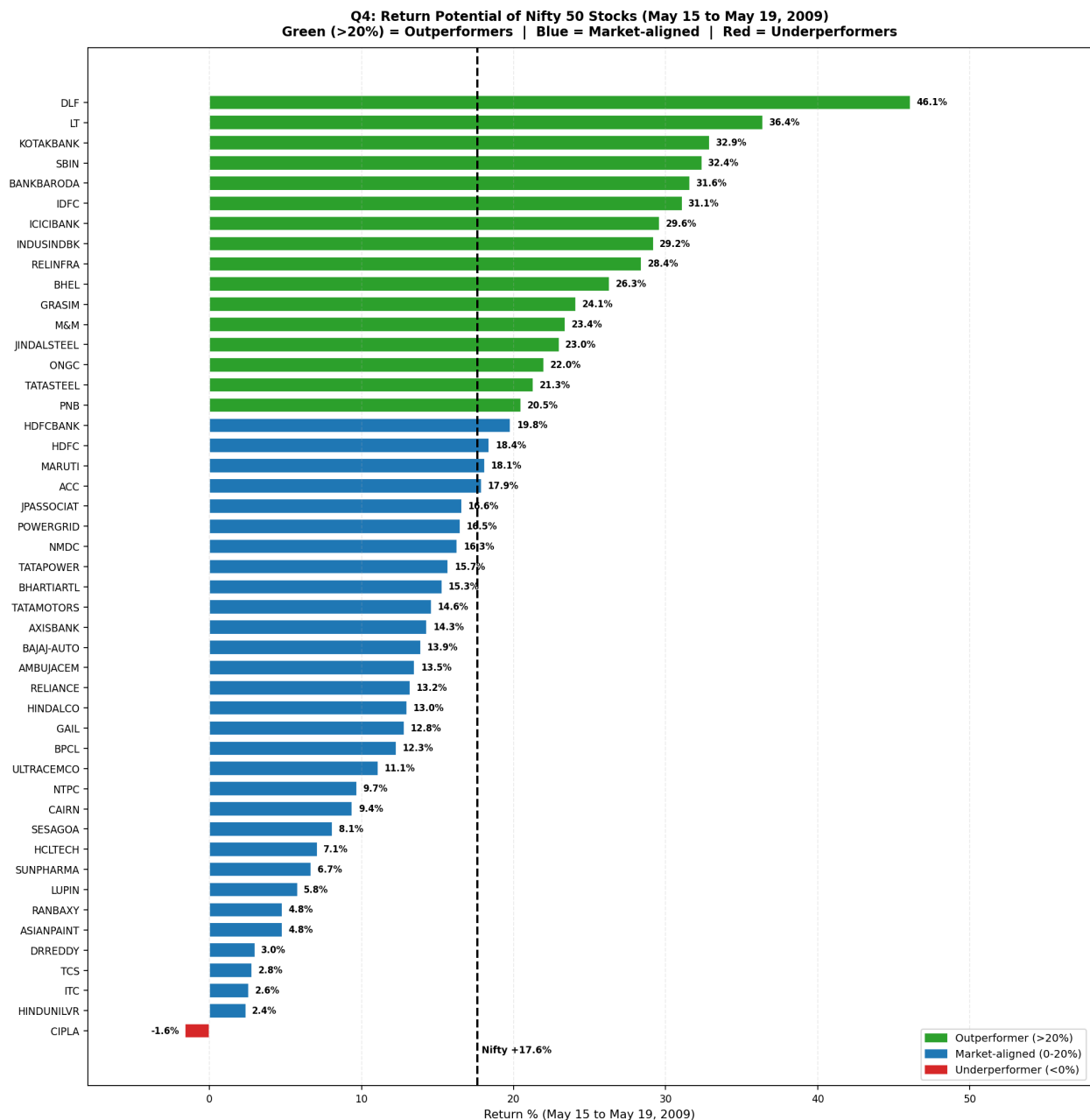


Figure 7: Return (%) for each Nifty 50 stock from May 15 to May 19, 2009. Green bars cleared 20% (outperformers), blue bars tracked the market, red bar fell below zero. The dashed line marks the Nifty benchmark of +17.6%.

4.1 Sector breakdown

Sector	Key Stocks	Avg Return	What drove it
Construction / Infra	DLF (46.1%), LT (36.4%), JP Assoc (16.6%)	~33%	Stable government = infra budgets get passed and projects move forward.
PSU and Private Banks	KOTAKBANK (32.9%), SBIN (32.4%), BANKBARODA (31.6%), ICICIBANK (29.6%)	~31%	Clear mandate = policy continuity, stable rates, easier credit. PSU banks saw heavy short-covering.
Power / Capital Goods	BHEL (26.3%), RELINFRA (28.4%), POWERGRID (16.5%)	~24%	Govt capacity-expansion plans benefit equipment makers and grid operators.
Metals / Steel	TATASTEEL (21.3%), JINDALSTEEL (23%)	~22%	Cyclical rally when infrastructure spending is expected to increase.
Automobiles	BAJAJ-AUTO (13.9%), TATAMOTORS (14.6%), M&M; (23.4%)	~17%	In line with market. Stable domestic demand; no direct policy angle.
IT / Software	TCS (2.8%), HCLTECH (7.1%)	~5%	Rupee appreciated 4.5% vs dollar, directly denting dollar-denominated revenues.
Pharmaceuticals	CIPLA (-1.6%), DRREDDY (3%), RANBAXY (4.8%)	~2%	Defensive sector loses appeal in a risk-on rally. Funds rotated into cyclicals.
FMCG / Diversified	HINDUNILVR (2.4%), ITC (2.6%)	~2.5%	Classic defensives get sold when investors shift to higher-beta opportunities.

4.2 Top 5 and bottom 1

Ran k	Stock	Sector	May 15 (Rs.)	May 19 (Rs.)	Return
1	DLF	Construction	258.25	377.40	46.1%
2	LT	Engineering	982.50	1,339.80	36.4%
3	KOTAKBANK	Banking	502.00	667.00	32.9%
4	SBIN	Banking (PSU)	1,319.10	1,745.85	32.4%
5	BANKBARODA	Banking (PSU)	342.00	450.00	31.6%
--	CIPLA	Pharma	230.00	226.40	-1.6%

4.3 Why stocks diverge from the index

- **Sector rotation:** Decisive election results shift capital from defensives (pharma, FMCG, IT) into cyclicals (infra, banks, metals). This is why CIPLA fell while DLF nearly doubled the index return.
- **Beta differences:** DLF beta ~1.8-2.0; CIPLA beta ~0.4-0.5. High-beta stocks amplify index moves in both directions.
- **Policy linkage:** Infrastructure and banking earnings are directly tied to government spending, interest rates, and public investment -- all of which improve with a stable mandate.

- **Currency move:** The rupee gained 4.51% in two days (Rs.49.55 to Rs.47.49). TCS and HCLTECH earn in dollars -- rupee appreciation reduces their reported earnings.
- **Short covering:** PSU bank stocks were heavily shorted pre-election. Once the result was clear, forced short-covering amplified gains beyond fundamentals.
- **Circuit breakers:** Only 202 of ~1,500 NSE stocks traded on May 18. Many recorded returns were nominal prices at upper circuits, not actual traded prices.

Question 5: Options Strategies for Lower Circuit / High Volatility

The 2009 outcome was about as good as the market could have hoped for. Sharma also had to think through what would happen if it went the other way -- a result with six or seven parties trying to stitch together a coalition would almost certainly hit the market hard, possibly triggering lower circuits. The nearest reference point is January 2008, when Nifty shed over 26% in under a week across two lower circuit breaches. The strategies here use the same Exhibit 3 prices from May 15, built around that bearish or high-volatility scenario.

5.1 The January 2008 crash as a reference

Date	Open	High	Low	Close	What happened
16 Jan 2008	6,065	6,065	5,825.75	5,935.75	Market at peak
21 Jan 2008	5,705	5,705	4,977.10	5,208.80	First big gap down
22 Jan 2008	5,203	5,203	4,448.50	4,899.30	Lower circuit hit
25 Jan 2008	5,035	5,399	5,035.05	5,383.35	Partial recovery
Peak to trough	--	--	--	--	~26.6% in 5 sessions

5.2 Strategies for a bearish or volatile outcome

#	Strategy	Type	What you buy / sell	Max Loss	When you profit
1	Long Put	Bearish	Buy Put K=3600 @ Rs.119.65	Rs.119.65	Nifty below 3,480.35
2	Bear Put Spread	Bearish / Defensive	Buy Put K=3700 @ Rs.162 Sell Put K=3500 @ Rs.89 Net = Rs.73	Rs.73	Nifty below 3,627 Full Rs.127 if Nifty falls to 3,500 or below
3	Long Straddle	Volatile	Buy Call K=3700 @ Rs.150 Buy Put K=3700 @ Rs.162 Total = Rs.312	Rs.312	Nifty above 4,012 or below 3,388
4	Long Strangle	Volatile / Cheaper	Buy Call K=3800 @ Rs.108 Buy Put K=3600 @ Rs.119.65 Total = Rs.227.65	Rs.227.65	Nifty above 4,028 or below 3,372

5.3 Payoff Formulas

Long Put : $\text{Max}(3600 - S_T, 0) - 119.65$

Bear Put Spread : $\text{Max}(3700 - S_T, 0) - \text{Max}(3500 - S_T, 0) - 73$

Long Straddle : $\text{Max}(S_T - 3700, 0) + \text{Max}(3700 - S_T, 0) - 312$

Long Strangle : $\text{Max}(S_T - 3800, 0) + \text{Max}(3600 - S_T, 0) - 227.65$

5.4 Payoff Diagrams

Q5: Strategies for Nifty Lower Circuit / Heavy Plunge
Reference: Jan 2008 (Nifty fell ~26.6% in 5 sessions)

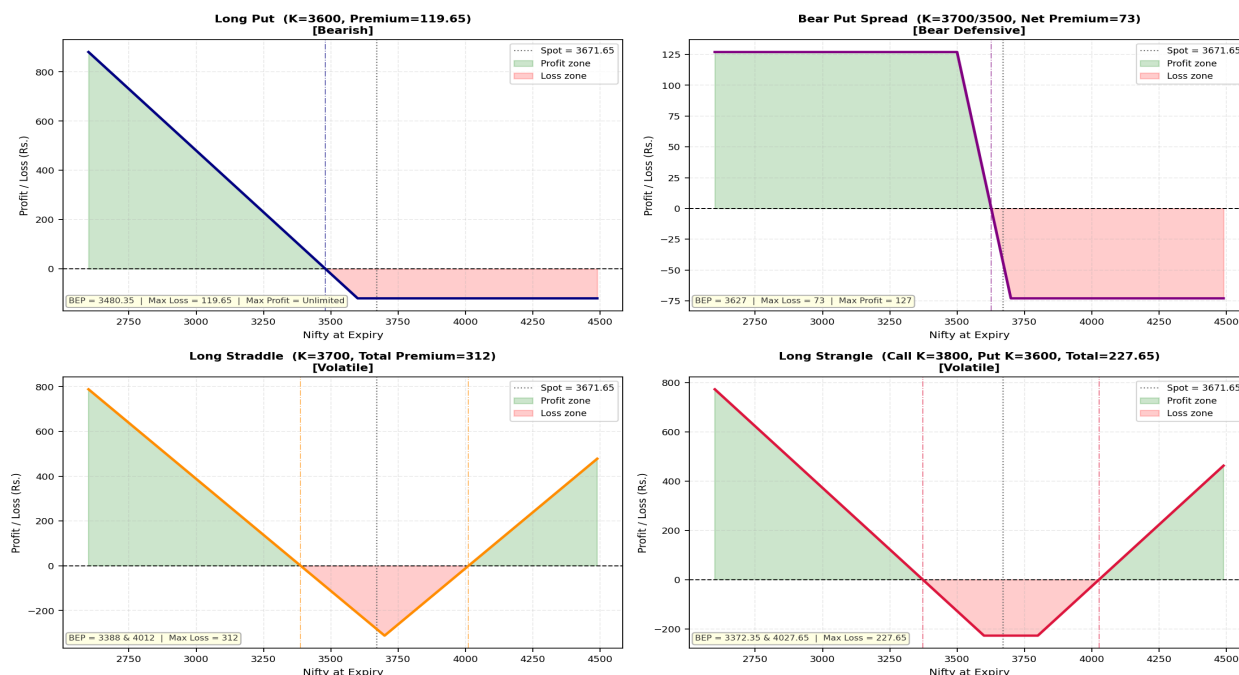


Figure 8: Payoff diagrams for all four bearish / volatile strategies. The dotted grey line shows the May 15 spot of 3,671.65. Dash-dot lines mark the break-even points for each strategy.

5.5 Break-even summary

Strategy	Break-even level(s)	Max loss	Max profit
Long Put (K=3600)	Rs.3,480.35 (Nifty must fall below this)	Rs.119.65	Unlimited -- grows as Nifty falls
Bear Put Spread (3700/3500)	Rs.3,627 (below)	Rs.73	Rs.127 if Nifty reaches 3,500 or lower
Long Straddle (K=3700)	Rs.3,388 downside Rs.4,012 upside	Rs.312	Unlimited in either direction
Long Strangle (3800C/3600P)	Rs.3,372 downside Rs.4,028 upside	Rs.227.65	Unlimited in either direction

5.6 Which strategy to pick and why

- **If you believe the third front will win and a sharp fall is coming:** go with the Bear Put Spread. It costs only Rs.73 and returns Rs.127 if the index drops even 5% from current levels. The maximum loss is capped and the break-even is close enough that a moderate dip already puts the trade in profit.
- **If you expect a big move but are not sure of direction:** the Long Strangle is the better choice over the Straddle. It costs Rs.84.35 less (Rs.227.65 vs Rs.312), and the break-even range is actually wider on both sides.
- **Why Sharma avoided naked futures:** A futures position has no floor on losses. If a short futures trade went wrong and the market gapped up 17% -- exactly what happened -- the

loss would be Rs.636 per lot with no way to limit it. Options always define the worst-case loss upfront.

- **The 2004 comparison:** After the 2004 verdict the market fell roughly 20% over two sessions. A put bought 5% out of the money ahead of that result would have returned over 1,000%. Holding even a small put position going into a binary event like an election has historically paid off well, even when the premium felt expensive at the time.